Anantha Krishnan Orunnukaran Mani

Mechanical and Aerospace Engineer | Applied Mathematics

Trieste, Italy

+3 9 3 6 6 8 2 2 3 2 0 9 — ananthakrishnanom [at] gmail.com

github.com/ananthkrishna96 — gitlab.com/ananth96

linkedin.com/in/au4702 — ananthkrishna96.github.io

PROFESSIONAL PROFILE

Computational mechanics and scientific computing researcher specializing in reduced-order modeling, scientific machine learning, and data-driven surrogate modeling for multiphysics simulations. Experienced in developing numerical methods and scientific software to accelerate high-fidelity CFD and FEA workflows using projection-based, intrusive, and non-intrusive ROM techniques, including neural-network surrogates and autoencoder latent models. Research focuses on parametric PDE systems, thermomechanical modeling of plates and shells, uncertainty quantification, optimization, and engineering digital twins for real-time simulation.

TECHNICAL EXPERTISE

Engineering Design	GeoCreate, FreeCAD, ANSYS SpaceClaim, CATIA, SolidWorks	
CAE Simulation	*Fluids and Thermal:*	OpenFOAM, AVBP, ANSYS Workbench, COMSOL Multiphysics
	Structural Mechanics:	Abaqus FEA, Altair HyperWorks, HyperMesh, Mechanical APDL
	Other Tools:	CENTAUR, h!p, Gmsh, deal.II, FEniCS, FreeFEM++
ROM & HPC	*Key Methods:*	Projection-based ROM, Intrusive and non-intrusive ROM, Artificial Neural Networks, Autoencoders, Kernel PCA, System Identification, Data-driven modeling, Parametric optimization, Vectorization, Parallel computing
	Tools and Libraries:	RBniCS, EZyRB, Numba, mpi4py, PyTorch, Scikit-learn
Software Development	*Programming:*	Python, C++, MATLAB
	Development:	Version control, Git, Collaborative development, Code review, CI/CD
Communication Skills	*Documentation:*	Microsoft Office, LibreOffice, Inkscape, LaTeX
	Languages:	English (Fluent), Malayalam (Native), Hindi (Native), Italian (Beginner)

EDUCATION

10/2022 – Present

PhD Researcher in Mathematical Analysis, Modeling and Applications

Scuola Internazionale Superiore di Studi Avanzati (SISSA), Trieste, Italy

- High-fidelity fourth-order finite element models and POD-based ROMs for orthotropic plate and shell systems on elastic foundations, with unilateral contact and panel-interface coupling.
- Intrusive POD–Galerkin and data-driven ROMs using POD–NN, neural-network surrogates, autoencoder latent models, and ML-based coefficient prediction for fast many-query simulation.
- Thermomechanical and transient dynamic extensions for structural response prediction, coupled simulation, verification, and ROM-oriented workflows.
- *Collaborations*: NextGenerationEU–PNRR/EPS Italia Srl, SISSA mathLab, Univ. of Houston.

09/2019 – 07/2022

Master's Degree in Space and Astronautical Engineering

Sapienza University of Rome, Italy.

Specialization: Fluid Dynamics, Structural Mechanics, Thermo-acoustics, and Control Systems.

Projects:

- *Master Thesis:* Characterization of the aeroacoustic transfer matrix of narrow ducts and perforated plates based on compressible Computational Fluid Dynamics and System Identification.
  - Data analysis and Modeling of Turbulent Flows, TU Berlin, Germany.
  - Thesis DOI: `https://doi.org/10.6084/m9.figshare.20210714.v1`
- Design and analysis of the 6U HyonoSat CubeSat - OBDH & Thermal Subsystem.

06/2014 – 05/2018

Bachelor of Technology in Aerospace Engineering

SRM Institute of Science and Technology, Chennai, India.

Projects:

- Design and fabrication of surveillance Hovercraft.
- Remote controlled weather monitoring and surveillance UAV.
- Free Convection heat transfer analysis of Thermal Flask.

PROFESSIONAL EXPERIENCE

09/2021 – 05/2022

Research Trainee

Institute of Fluid Dynamics and Technical Acoustics, **Technical University of Berlin, Germany**

Research Group: *Data Analysis and Modeling of Turbulent Flows*

- Performed CAD modeling (GeoCreate) and mesh generation (CENTAUR) for CFD simulations.
- Conducted CFD analysis of steady and acoustically excited duct/pipe flows using the AVBP solver with broadband forcing and boundary treatments.
- Developed low-order acoustic network models via CFD/System Identification to improve computational efficiency (MATLAB).
- Performed post-processing, analysis and technical reporting of simulation results.
- *Skills:* CAD, CFD, ROM, System Identification, Data Analysis, Machine Learning.

07/2018 – 01/2019

Project Intern

Techzilon Training Solutions, Bangalore, India

- Completed training in CFD and Finite Element Analysis for engineering simulations.
- Performed CAD modeling (CATIA V5) and mesh generation (HyperMesh, ANSYS Mesh).
- Conducted structural FEA using Optistruct, RADIOSS and ANSYS Mechanical.
- Performed CFD analysis of airfoil flows using ANSYS CFX and Fluent.
- *Project:* Modal analysis and FE modeling of a car door trim sub-assembly.
- *Skills:* CAD clean-up, grid convergence studies, technical reporting.

Engineering simulation and numerical modeling for industrial clients

Research Intern (Portfolio Project)

- Developed physics-based solar-thermal models within the SunPeek D-CAT framework, including collector-field dynamics, 1D pipe heat-loss, and lumped 0D heat-exchanger modeling.
- Implemented coupled forward simulations for collector fields, piping, and heat exchangers using ODE-based scientific computing methods.
- Conducted validation and yield analysis of solar-thermal plants using real operational data, sensor mapping, automated tests, and diagnostic plots.
- *Skills:* Scientific software development, 0D/1D dynamic simulation, numerical algorithms, physics-based modeling.

Scientific Computing Algorithms and Software Developer (Portfolio Project)

- Performed FEA and CFD modeling for multi-scale coupled-physics simulations in additive manufacturing of mechanical and electronic systems.
- Implemented multi-objective parametric optimization algorithms to improve simulation efficiency and design exploration.
- Integrated advanced numerical models into simulation codebases for engineering analysis and industrial applications.
- *Skills:* Numerical algorithms development, CFD/FEA, Parametric Optimization, C++.

CONFERENCES AND SCHOOLS

09/2025 SIMAI 2025 – Italian Society of Applied and Industrial Mathematics Conference, Trieste, Italy

- Contributed talk: *C^0-DG Solver and Reduction Strategies for Orthotropic Plates on Elastic Foundations with Interface Contacts*.

07/2025 SAMM 2025 – 10th GAMM Juniors Summer School on Model Order Reduction in Mechanics

06/2023 Dobbiaco Summer School 2023 – Data-driven Methods for the Computational Sciences

CERTIFICATIONS AND TRAINING

01/2019 Master Diploma in Computational Fluid Dynamics (CFD) and Computer Aided Engineering (CAE)

06/2017 Bharat Heavy Electricals Limited (BHEL), Hyderabad, India – Industrial Training

06/2016 Taneja Aerospace and Aviation Limited, Tamil Nadu, India – Industrial Training

MOOCs

- Google IT Automation with Python – Professional Certificate
- Python Data Structures – University of Michigan
- MATLAB Programming for Engineers – Coursovie

SERVICE / LEADERSHIP

10/2022 – Present Member – INdAM GNCS (Gruppo Nazionale per il Calcolo Scientifico)

02/2023 – 10/2025 Webmaster – Mathematics Area, SISSA

09/2023 – 04/2024 Treasurer – SIAM Student Chapter, SISSA

09/2021 – 02/2022 Marketing and Incoming Global Volunteer – AIESEC Italia

02/2020 – 12/2021 Student Tutor – Sapienza University of Rome